

ESSEX COUNTY AP, NJ, USA

WMO: 724094

Lat: 40.876N

Lon: 74.283W

Elev: 173

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54743

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.2	(c) 13.4	(d) -8.0	(e) 3.6	(f) 12.3	(g) -3.5	(h) 4.6	(i) 17.4	(j) 21.2	(k) 29.0	(l) 19.7	(m) 27.6	(n) 6.3	(o) 320	(p) 0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	92.2	73.7	89.8	72.5	87.0	71.4	76.9	86.4	75.4	83.8	74.2	82.2	7.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.2	128.7	80.4	72.9	123.0	79.1	72.0	119.1	78.3	40.2	86.6	38.9	83.9	37.7	81.8	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	16.1	13.1	4.1	96.8	5.0	2.9	0.5	99.0	-2.4	100.7	-5.2	102.3	-8.8	104.5
			2.6	79.2	4.7	1.9	-0.8	80.6	-3.5	81.7	-6.1	82.8	-9.5	84.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	30.0	32.4	40.4	51.8	61.8	70.5	75.6	73.9	66.9	55.1	44.8	35.6
	DBStd	17.70	9.92	8.99	9.00	8.68	7.78	6.44	5.38	5.30	6.63	8.05	8.26	8.61
	HDD50	2211	623	495	321	76	3	0	0	0	0	40	198	455
	HDD65	5248	1086	912	764	407	159	22	1	3	55	320	606	913
	CDD50	3433	2	3	23	130	368	614	795	742	508	199	42	7
	CDD65	995	0	0	0	11	59	187	331	280	113	14	0	0
	CDH74	8962	0	1	5	175	619	1704	3162	2374	836	84	2	0
	CDH80	3161	0	0	0	52	201	586	1256	833	222	11	0	0
Wind	WSAvg	4.8	5.7	5.8	6.3	6.1	4.7	4.2	3.7	3.3	3.6	4.2	4.6	5.3
Precipitation	PrecAvg	47.6	3.5	2.9	4.1	3.8	3.7	4.3	4.3	4.6	4.0	3.9	3.7	4.0
	PrecMax	66.1	7.4	6.2	10.4	9.9	9.0	9.8	8.2	13.7	8.3	13.6	6.5	7.3
	PrecMin	34.0	1.7	0.8	0.7	1.6	0.7	1.1	1.5	1.7	1.0	0.0	1.2	1.1
	PrecStd	8.4	1.4	1.3	2.1	1.9	1.6	2.2	1.7	2.6	2.2	3.1	1.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.6	64.2	73.1	86.6	90.6	93.5	97.0	95.3	90.8	81.3	71.7	65.5
		MCWB	56.7	55.0	58.2	64.8	69.5	73.1	74.7	76.2	73.6	68.1	62.2	58.1
	2%	DB	55.0	56.6	65.8	79.0	85.7	89.9	92.9	90.6	85.7	75.7	66.5	58.1
		MCWB	50.0	48.4	53.7	60.7	67.9	72.6	74.2	73.8	70.5	65.6	58.8	53.7
	5%	DB	48.1	51.2	60.1	72.8	81.2	86.3	90.1	87.8	81.9	72.3	62.9	53.3
		MCWB	42.8	43.3	49.3	58.0	65.6	70.9	73.0	72.1	68.8	63.9	54.9	48.5
	10%	DB	43.9	45.6	54.6	67.2	76.3	83.1	87.0	84.4	79.0	68.9	58.6	48.7
		MCWB	39.0	39.3	46.2	54.1	63.2	68.6	72.0	70.5	67.9	61.2	52.0	43.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.4	58.8	60.4	68.1	73.0	76.2	79.2	79.3	76.5	72.6	64.8	60.4
		MCDB	60.8	63.5	69.7	79.1	84.8	87.1	91.4	90.3	85.0	76.3	69.3	63.1
	2%	WB	50.8	50.4	55.9	63.8	70.5	74.8	77.2	76.6	74.4	69.2	60.5	54.7
		MCDB	54.2	55.8	63.5	74.5	81.1	85.2	87.4	85.1	80.5	73.9	64.8	57.4
	5%	WB	44.1	44.0	51.2	60.4	68.5	73.2	75.7	75.1	72.8	65.9	57.1	49.8
		MCDB	47.2	48.7	57.5	70.4	77.3	82.7	84.3	82.6	78.3	70.6	61.0	52.6
	10%	WB	39.4	39.9	47.0	56.8	66.1	71.4	74.4	73.6	70.7	62.6	53.3	44.5
		MCDB	43.4	45.4	54.0	64.3	73.7	79.8	82.8	80.5	76.3	66.8	57.1	47.4
Mean Daily Temperature Range	5% DB	MDBR	14.9	16.7	17.6	19.8	19.9	18.8	18.6	18.1	18.6	18.5	17.4	14.2
		MCDBR	20.8	24.5	26.1	28.7	25.9	22.4	21.7	20.8	22.0	22.6	22.4	19.8
		MCWBR	17.2	18.0	16.3	14.2	11.5	9.6	7.7	7.6	10.0	12.6	15.1	17.2
	5% WB	MCDBR	19.6	22.3	22.7	24.5	22.2	19.3	17.6	17.4	17.0	17.7	18.2	18.2
		MCWBR	17.4	18.1	16.7	14.5	11.3	9.7	7.6	7.7	9.5	12.3	15.2	17.6
Clear-Sky Solar Irradiance	taub	0.309	0.324	0.357	0.379	0.425	0.487	0.488	0.474	0.397	0.359	0.327	0.309	
	taud	2.448	2.406	2.387	2.363	2.262	2.126	2.136	2.167	2.354	2.470	2.518	2.515	
	Ebn at Noon	269	281	283	283	271	254	252	252	266	266	261	261	
	Edh at Noon	25	30	34	37	43	49	48	45	35	28	23	22	
All-Sky Solar Radiation	RadAvg	584	869	1169	1500	1665	1823	1872	1620	1354	935	656	493	
	RadStd	48	88	95	149	174	167	95	106	141	97	59	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+302	+202
Regional (63 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	+121

Nomenclature: See separate page